

Topic 5: Anti-Forensics & Cryptography

F20FO/F21FO – Digital Forensics

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Previous Topic

- Topic 4: Evidence Types & Collection
 - Files and file signatures
 - File signature analysis

Today's topic overview

- Topic 5: Anti-Forensics & Cryptography
 - Introduction
 - Steganography
 - Cryptography
 - Implications for Digital Forensics

Introduction

Introduction

- There are several techniques that may *incidentally* be used that have a negative impact on digital forensics
- **Incidentally:** meaning that the resulting impacts on the data was not intended to negatively impact forensics data
- **Negative impact:** meaning that the use of the technique has made it more difficult to recover or analyse the data

Introduction

Some incidental techniques include

- Even wear levelling on flash drives. We discussed earlier how this storage architecture choice negatively impacts digital forensics
- Encrypting your storage data. While this might be not to protect one's data from compromise by an attacker, it also makes data recovery more difficult for digital forensics
- De-fragmenting a drive might overwrite previously unallocated clusters that may otherwise have contained useful forensics data
- Using advanced data deletion on a storage device by default, as they become more readily available to support the protection of personal privacy

Introduction

- Thankfully, for the DF expert, not many/all users make use of the previous practices (though usage is increasing)
- However, a criminal that anticipates that their data might be used against them may proactively try to make life more challenging for a DF expert by
 - Encrypting their storage data.
 - Periodically de-fragmenting their storage drives
 - Permanently deleting old data, i.e., overwriting the data on storage, rather than simply deleting (“deallocating”) the data from a file allocation

Introduction

- We'll first review some techniques for hiding data on a storage device using steganography and/or cryptography
- While computer security applications almost exclusively using modern cryptographic techniques, as would be covered in a computer security class, criminals still rely on older techniques since tools are readily available and can be easier to understand
 - Steganography
 - Historical cryptography

Steganography

Steganography

- Steganography - “hidden writing”
- Steganography is defined as a technique for hiding messages and data within a multimedia file
- The secret message embedded into the multimedia file as a cover and can be transmitted using emails or other media
- The message can then be extracted at the destination to avoid violation on confidentiality of data
- Multimedia files includes:
 - Images
 - Video
 - Audio

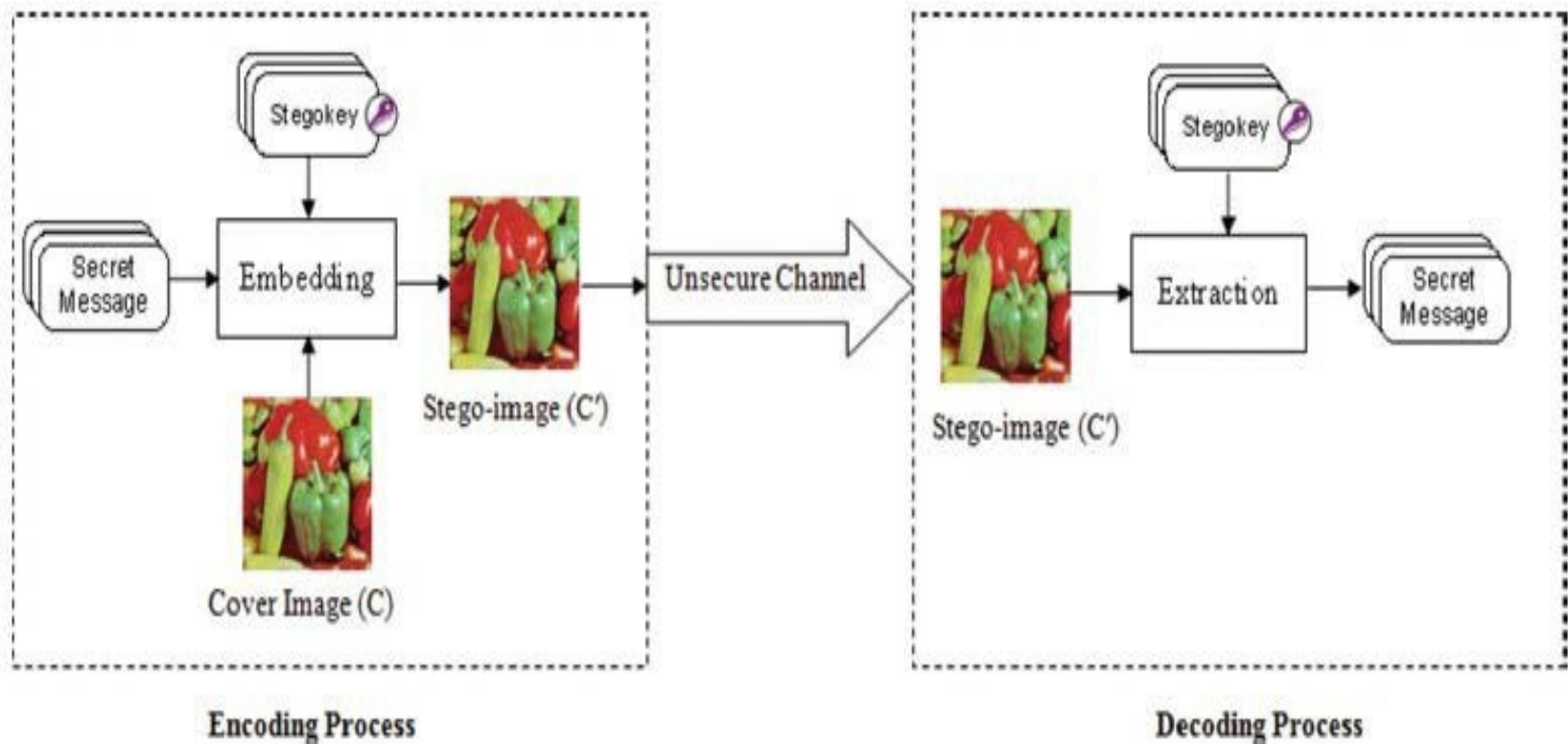
Steganography is not new

- **Herodotus** (485 – 525 BC) is the first Greek historian
 - The story of the war between the huge Persian empire and the much smaller Greek city-states
 - Histiaeus, who wanted to encourage Aristagoras of Miletus to revolt against the Persian king
 - To securely convey his plan, Histiaeus shaved the head of his messenger, wrote the message on his scalp, then waited for the hair to regrow. The messenger, apparently carrying nothing contentious, could travel freely. Arriving at his destination, he shaved his head and pointed it at the recipient

Steganography is not new

- **Pliny the Elder** explained how the milk of the thithymallus plant dried to transparency when applied to paper but darkened to brown when subsequently heated, thus recording one of the earliest recipes for invisible ink
- **Johannes Trithemius** (1518) wrote the first printed book on cryptology. He invented a steganographic cipher in which each letter was represented as a word taken from a succession of columns. The resulting series of words would be a legitimate prayer
- **Giovanni Battista Porta** (around 1560) described how to conceal a message within a hard-boiled egg by writing on the shell with a special ink made with an ounce of alum and a pint of vinegar. The solution penetrates the porous shell, leaving no visible trace, but the message is stained on the surface of the hardened egg albumen, so it can be read when the shell is removed

Steganography in practice



- **Masking and Filtering**
 - Information is hidden inside of an image using digital watermarks that include information such as copyright, ownership, or licenses
 - The purpose is different from traditional steganography since it is adding an attribute to the cover image thus extending the amount of information presented
- **Algorithms and Transformations**
 - Data is hidden in mathematical functions that are often used in compression algorithms
 - The idea of this method is to hide the secret message in data bits in the least significant coefficients
- **Least Significant Bit Insertion**
 - The most common and popular method of modern-day steganography is to make use of the LSB of a picture's pixel information. Thus, the overall image distortion is kept to a minimum while the message is spaced out over the pixels in the images
 - This technique works best when the image file is larger than the message file and if the image is grayscale

Application



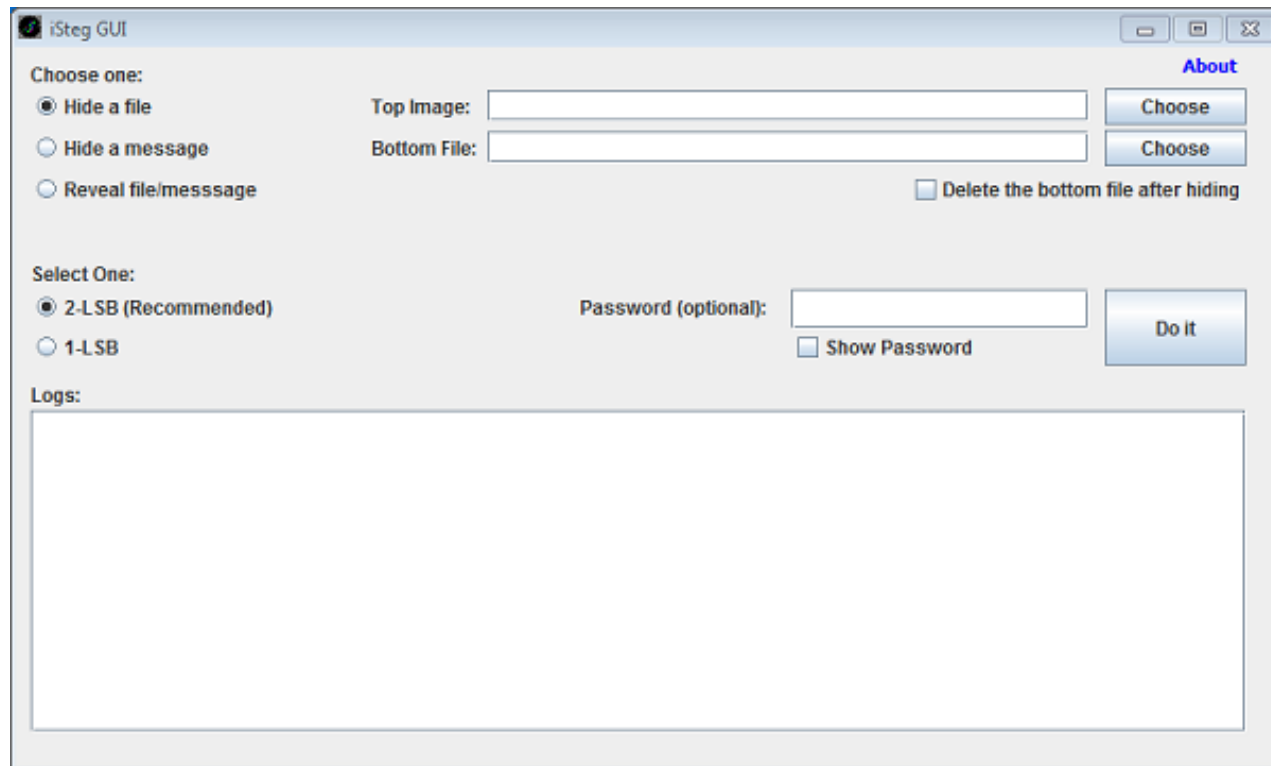
Steganography Tools

- Some tools
 - iSteg
 - Steghide
 - Stegosuite
 - Hermetic Stego 9.37
 - Outguess
 - EzStego

Image Steganography

- Hide file in image files for most formats
 - PNG
 - JPEG
 - BMP, etc.
- Hidden data cannot be detected visually
- Techniques used:
 - Masking and Filtering
 - Algorithms and Transformations
 - Least Significant Bit Insertion

- Hide multiple application in a single image
- Uses LSB method



Audio Steganography

- Hide file in Audio for different formats
 - MP3
 - WAV, etc.
- Techniques used:
 - Algorithms and Transformations
 - Least Significant Bit Insertion

- Hides data in .WAV files
- Python based
- Command line tool

Run WavSteg with the following command line arguments:

Command Line Arguments:

-h, --hide	To hide data in a sound file
-r, --recover	To recover data from a sound file
-i, --input TEXT	Path to a .wav file
-s, --secret TEXT	Path to a file to hide in the sound file
-o, --output TEXT	Path to an output file
-n, --lsb-count INTEGER	How many LSBs to use [default: 2]
-b, --bytes INTEGER	How many bytes to recover from the sound file
--help	Show this message and exit.

Example:

```
$ stegolsb wavsteg -h -i sound.wav -s file.txt -o sound_steg.wav -n 1  
# OR  
$ stegolsb wavsteg -r -i sound_steg.wav -o output.txt -n 1 -b 1000
```

<https://github.com/ragibson/Steganography?tab=readme-ov-file#wavsteg>

Video Steganography

- Hide file in video formats
 - AVI
 - MPG4
 - WMV, etc.
- Both techniques in audio and image can be used
 - Video consists of audio and image
 - Many secret messages can be hidden in video as they are moving stream of audio and images

MSU Stego Video

- Can compress video
- There may be small distortions in video after information is hidden
- Can use passcode



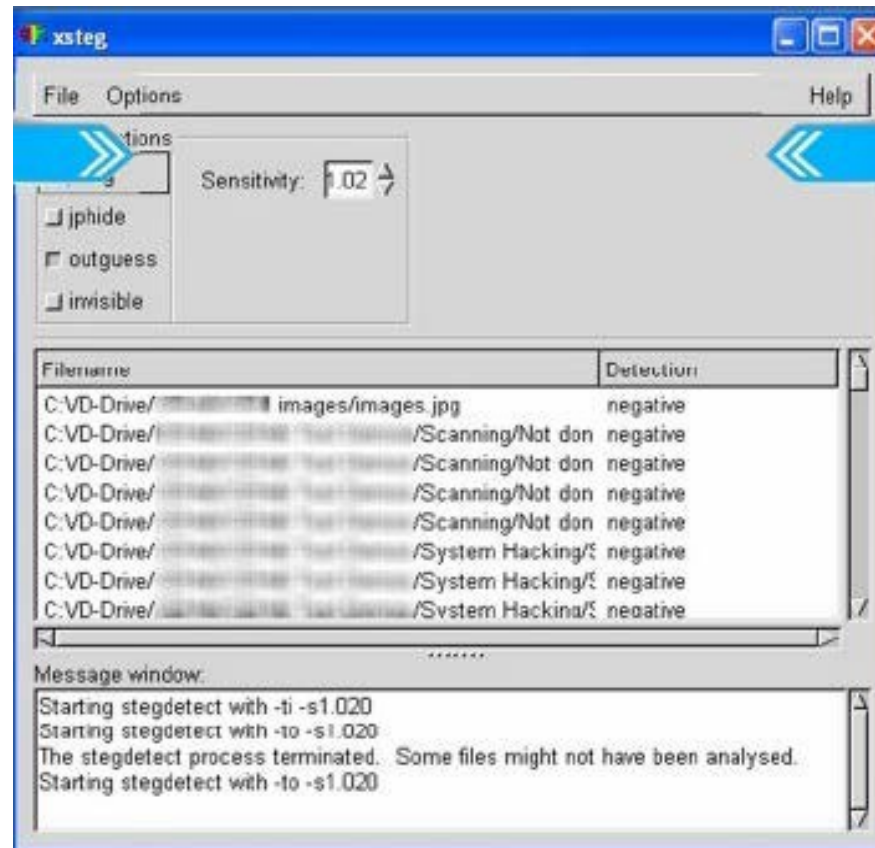
https://www.compression.ru/video/stego_video/index_en.html

Steganalysis

- The detection of steganography is called Steganalysis
- Visual Analysis tries to reveal the presence of secret communication through inspection, either with the naked eye or with the assistance of a computer
- Statistical (Algorithmic) Analysis reveals tiny alterations in an image's statistical behaviour caused by steganographic embedding
 - The nominally universal methods developed to detect embedded stego-data are generally less effective than the steganalytic methods aimed at specific types of embedding

Stegdetect

- Can detect use of steganography



Cryptography

Cryptography

- Cryptography is a Greek word meaning “secret writing”
- Cryptography is the science of secret, or hidden, writing
- Modern crypto employs complex algorithms to achieve this:
 - Our emphasis is on implementation and system security (not the math itself, covered in F20CN/F21CN)

Two main Components

- **Encryption**
 - Data/message hiding, so that they cannot be read by anyone other than the intended recipient
- **Authentication & Integrity**
 - Ensuring that users of data/resources are the persons they claim to be and that a message has not been altered

Terminology

- **plaintext** - original message
- **ciphertext** - coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering ciphertext from plaintext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis** (codebreaking) - study of principles/methods of deciphering ciphertext without knowing key
- **cryptology** - field of both cryptography and cryptanalysis

Applied Cryptography

- Maintains data security in an insecure environment
- Can provide:
 - Confidentiality
 - Integrity
 - Authentication (no one can encrypt without the key)
 - Non-repudiation

Example Use

- Cryptography used in:
 - VPNs
 - Storing hashes of passwords
 - Internet banking
 - Messaging Apps
 - WhatsApp, Telegram
- Where is Cryptography also used?

Encryption Terminology

- Encryption algorithms are standardised & published
- The key which is an input to the algorithm is secret
- Key is a string of numbers or characters
- If same key is used for encryption & decryption, the algorithm is called **symmetric**
- If different keys are used for encryption & decryption, the algorithm is called **asymmetric**

Cryptanalysis

- Cryptanalysis is the process of breaking an encryption code
 - Non-trivial process
- Several techniques can be used to deduce the algorithm
 - Attempt to recognise patterns in encrypted messages
 - to be able to break subsequent ones by applying a straightforward decryption algorithm

Cryptanalysis

- Attempt to infer some meaning without even breaking the encryption, such as noticing an unusual frequency of data (e.g. words, text)
- Attempt to find weaknesses in the implementation or environment of use of encryption or in the encryption algorithm

Types of Attacks

- Ciphertext only:
 - The attacker only has the ciphertext, attempts to get the plaintext and/or key
- Known plaintext:
 - Attacker has the plaintext and ciphertext, attempts to determine the key
- Chosen plaintext:
 - Attacker specifies plaintext and receives ciphertext, attempts to find the key

Symmetric Encryption

- Algorithms in which the same key is used for encryption and decryption are **Symmetric**
 - E.g. Caesar Cipher

Symmetric Encryption Types:

- **Block Ciphers**
 - Encrypt data one block at a time (typically 64 bits, or 128 bits)
 - Used for a single message
- **Stream Ciphers**
 - Encrypt data one bit or one byte at a time
 - Used if data is a constant stream of information

Advantages of Symmetric Encryption

- Strength of algorithm is determined by the size of the key
 - The longer the key the more difficult it is to crack
- Key length is expressed in bits
 - Typical key sizes vary between 48 bits and 448 bits
 - For 64-bit key there are 2^{64} possible keys
 - For 128-bit key there are 2^{128} possible keys
 - Each additional bit added to the key length doubles the security
- To crack the key, the hacker must use brute-force (i.e. try all the possible keys till a key that works is found)
 - Supercomputer can crack a 56-bit key in 24 hours
 - It will take 2^{72} times longer to crack a 128-bit key

4,722,366,482,869,645,213,696 days

Cryptography Techniques

- Transposition

- Rearranges the plaintext characters, without changing them, to produce ciphertext
- For example, columnar transposition rearranges the plaintext into columns
- For example, for plaintext “HELLO THERE”, rearrange into **four** columns:

HELL

O TH

ERE

- Read down, ciphertext = “HOE E R LTE LH”
- Adjacent-character analysis does not help

Cryptography Techniques

- Substitution

- Changes the characters of the plaintext to produce the ciphertext

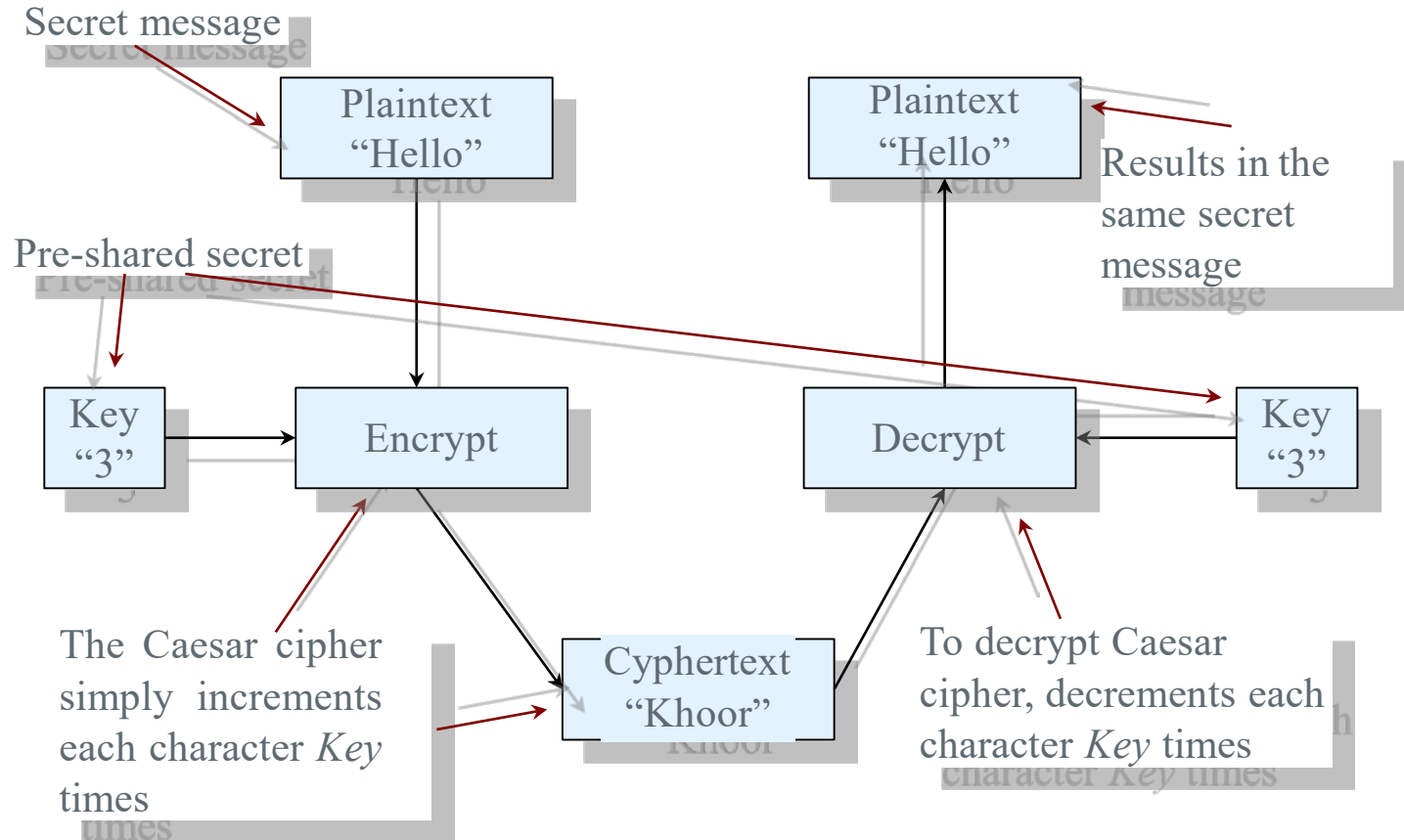
simple substitution: a secret table of characters to other characters

a=q, s=w, d=e...

Caesar Cipher

- The Caesar cipher is a simple substitution cipher
 - Changes characters
 - Shifts each character Key times
 - Julius Caesar always shifted by 3 places (so it was essentially keyless)
- It is easy to break
 - Character frequencies and other language based techniques
 - Only need to try 26 keys (or 27 if including spaces in the encoding)

Caesar Cipher



Caesar Cipher

- Caesar Cipher is a method in which each letter in the alphabet is rotated by “n” letters as shown (n was initially set to 3)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Let's encrypt, “Heriot”

KHULRW

Vigenère Cipher

- Assign password
- Repeat the password over the plain text

Message: DIGITAL, Password: CAT

Password: CATCATC

Ciphertext: FIZKTTN

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X

Vigenère Cipher

- Breaking the Vigenère cipher
 - Because the key repeats, if we know the length of the key we can compare every key-length characters
 - For example, if the key is 5 characters long, then we could look at every fifth letter
 - They are all encrypted with the same key character
 - Frequency analysis can be used to reveal which one is “e”, which can tell us the key...

Key Space

- The key space is the number of possible keys
- Various algorithms have a predefined key space
- Caesar cipher has 26 possible keys (extremely small key space)
- For a randomly chosen key, an attacker would on average need to search half of the key space (13 keys)
- Say for example, that it took 1 minute to manually try each decryption, it would take on average 13 minutes to “brute force” (try each possible key)

Key Space

- Caesar cipher is close to useless in terms of modern security
- It essentially relies on security through obscurity: that you don't know how the algorithm works

Keyless Algorithm

- An example of a keyless algorithm is **ROT13**
- It is a simple substitution cipher
- Increments each character by 13 (an example of the Caesar cipher)
- Used on online forums to hide spoilers, quiz answers, offensive jokes (and so on)

- Applying it again returns it to the encrypted state

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
NOPQRSTUVWXYZABCDEFGHIJKLMnopqrstuvwxyzabcdefghijklm

- ROT13 is not designed to provide actual secrecy
- ROT13 has (mistakenly) been used to create several insecure systems
- Used as part of a system to store Netscape Communicator email passwords

Breaking Substitution Ciphers

- Breaking a simple substitution cipher is much easier than bruteforce (guessing every key)
 - Ciphertext only attack
- Use the frequency of English letters
 - e: 12.7%, t: 9.1%, a: 8.1%
- Use the frequency of letter pairs
 - th, in, ...

Detecting Transposition and Substitution Ciphers

- Classical/historical ciphers can be detected as well, which can be important for digital forensics
- Letter frequencies (single, double, triple) can be helpful here
- For example
 - Output from a transposition cipher has the same single letter frequencies as the plaintext. But different double/triple letter frequencies
 - Output from a Caesar Cipher will have “masked” single, double and triple letter frequencies. In other words, the “shape” of the frequency graph is the same but the letters will be shifted
 - Other statistical techniques can be used for other ciphers, such as Vignere
 - Entropy, and similar measures (as discussed in and earlier lecture) can be used to detect ciphertext for modern ciphers (though more advanced techniques are needed to find or guess the key)

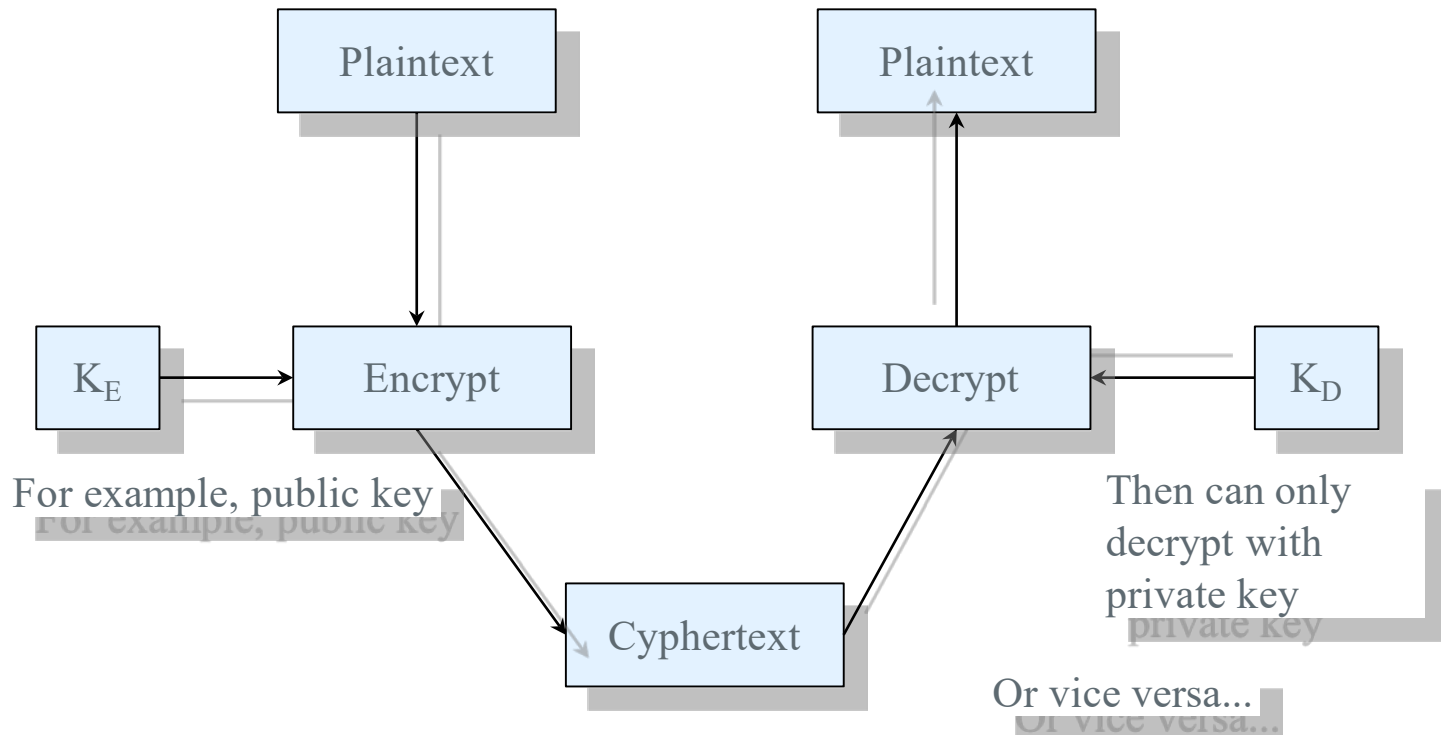
Key Distribution Problem

- **Symmetric key** cryptosystems use the same key to encrypt and decrypt
- Symmetric key cryptosystems rely on an out-of-band way of pre-sharing keys
- Requires a means of **key distribution**
- Need for new keys for each party communicated with
 - Alice, Bob, Carol, each require 2 keys (one for each other party they want to securely communicate with)

Asymmetric Crypto

- **Public key** (AKA asymmetric) cryptosystems use separate keys for encryption and decryption
- It is safe to tell anyone the encryption key, and only the person holding the decryption key can determine the original message

Public Key Crypto



Public Key

- Public keys can be made public:
- Used to encrypt
 - Email, web, ...
 - Discuss: what are the potential disadvantages of sharing keys via insecure channels? (Trust, MITM)
- Private keys must be kept secret (as with asymmetric keys)
- Used to decrypt
 - If a private key is known by a third party they can decrypt and modify any previous communications

Public key

- Some public key systems also work in reverse
 - Anything encrypted with the private key can only be decrypted with the public key
- Disadvantage of public key cryptography
 - can be 10 000 times slower than symmetric key
- Often used in protocols to exchange symmetric keys

Digital Signatures

- Digital signatures are an application of public key cryptography, to prove that a message:
 - Authentication: it was created by a known party
 - Integrity: that it has not been altered
 - Non-repudiation of origin: the signer cannot later deny signing the message
- Uses include:
 - Signing software
 - Legal digital documents

Digital Signatures

Analogy: a cheque

- The signature provides (some form of) authenticity
- Hard to alter (integrity)

- **Certificates**

How do we know who we are talking to?

- Discuss: someone calls you and claims to be representing the government
 - Do you trust them?
 - May depend on what the call is regarding
 - How can you verify? (Ask questions? Call them back?)

Certificates

- Often we don't know for sure
 - Often don't actually check signatures
 - May be vulnerable to attack
 - We all accept some risk in day-to-day life, in order to function
 - For example, when we buy food, how do we know our lunch has not been poisoned and our drinks not drugged?
- We establish some form of trust
- Trust through a respected third party
 - You might not know someone works for the same company as you, but you might know their manager and ask them
- Digital certificates can achieve this by establishing a hierarchy of trust (in identity)
 - One party can sign another parties key, showing that they trust in their identity

Certificates

- A certificate binds a public key with an identity
- A digital signature from a trusted third party authenticates the identity
- Public-key infrastructure (PKI) is a way of managing certificates
 - Web of trust
 - Certificate authorities

Web of Trust

- Alice can sign Bob's public key
- If Carol knows Alice is careful when signing, Carol can now trust Bob is who he says
- If Dave knows Carol, but no one else, then he can still trust the identity of Bob, if Carol has signed Alice's public key

Certificate Authorities

- Users rely on certificate authorities (CA) to issue certificates
- These trusted third parties (TTP) vet the identity of applicants
 - Typically, based on the fact that they can receive emails to the same domain (known as **domain validation**)

Certificate Authorities

- Centralised
 - Typically, everyone trusts certain CAs
 - Web browsers trust lots of CAs
- Hierarchical
 - Some CAs have the right to create other CA certificates
 - So to validate a certificate, you check that the public key has been signed by a CA that you trust (potentially with other CAs in between)

Certificate Authorities

- That is how HTTPS works
- When you go to <https://mybank.com>
- Your browser receives the certificate, and verifies it trusts it (it is signed by a known CA)
- It then uses the public key to encrypt messages to the site
 - only it can decrypt the traffic using the private key
- An attacker could attempt a man-in-the-middle attack, but unless the certificate they provide is also signed by a CA it will look suspicious

Weaknesses

- Not all CAs deserve to be trusted, they will give a certificate to just about anyone
- Some CAs belong to governments, so they can conduct MITM attacks against almost any site
- In 2001, VeriSign, one of the largest CAs, issued two Microsoft certificates to an unrelated person
- Comodo and DigiNotar have had CA security compromises

Implications for Digital Forensics

Reacting to Anti-Forensics

- How does a DF expert react to anti-forensics
- First, they must determine whether anti-forensics has been used
- We've already discussed some file analysis techniques
 - Analysing file content (**file signatures**), and not just relying on file extensions, to detect attempts to hide content
 - Using **Entropy** to detect encrypted content
- An investigation should also look for steganographic or encryption program files on the machine

Reacting to Anti-Forensics

- If encrypted content is found, then a first step is to find out what encryption might have been used
 - Cryptanalysis techniques for historical ciphers can be used, such as frequency analysis to both determine the cipher, and recover the plaintext
- If modern encryption techniques have been used, then one can search for the key
 - Live forensics can be used to search memory (if memory has been recovered)
 - Search files, emails, chat logs for mentions of keys or passwords (as many keys are derived from a user password)
- Use password cracking tools to attempt to decrypt the files

File Signatures for Encrypted Files

- Encrypted files, like other application files, have file signatures (“magic strings”) as discussed earlier
 - These “signatures” are very different from “digital signatures” used to provide cryptographic authenticity and integrity
- Consider the text file (‘plain’) containing the text “The quick brown fox jumped over the slow lazy dog.”
- The bytes for this file can be viewed with the `xxd` command:

```
$ xxd plain
00000000: 5468 6520 7175 6963 6b20 6272 6f77 6e20  The quick brown
00000010: 666f 7820 6a75 6d70 6564 206f 7665 7220  fox jumped over
00000020: 7468 6520 736c 6f77 206c 617a 7920 646f  the slow lazy do
00000030: 672e 0a                                g.
```

File Signatures for Encrypted Files

- For the text file, we can determine the file signature information using the 'file' command:

```
$ file plain
```

```
plain: ASCII text
```

- In this case, the file has no file signature and the OS will examine the file to determine that it contains ASCII text (with only printable characters, and not non-printable/binary characters)

File Signatures for Encrypted Files

- If we symmetrically encrypt this file with openssl (we'll be prompted for a password) using the encryption algorithm AES-256

```
$ openssl enc -aes256 -in plain -out  
ciphertext
```

```
$ file ciphertext
```

```
ciphertext: openssl enc'd data with  
salted password
```

- The 'file' command shows us new information
- If we look closely at the ciphertext file, we see the file signature

File Signatures for Encrypted Files

- File signature: 53 61 6c 74 65 64 5f 5f

```
$ xxd ciphertext
```

```
00000000: 5361 6c74 6564 5f5f 8ae9 4b9b 0d2a 9228  Salted__..K..*.(
00000010: 82bf 4dc5 38ad 0c67 12a1 3ddf 991b ae25  ..M.8..g..=....%
00000020: 6c39 b349 085d 3958 95f1 eae5 8bed e7f9  19.I.]9X.....
00000030: 7143 f5a8 b11a 0d8f 19f2 f819 4f86 4e1c  qC.....O.N.
00000040: 4582 c8f0 ad53 3e5b ebd2 dfe2 40b3 2ac3  E....S>[....@.*.
```

- We can see that the hex corresponds to the ascii text “Salted__” for the file signature for the openssl encrypted file
- Thus, a DF examiner could search for this file signature to find some encrypted files
- Other encrypted file formats will have different file signatures

File Entropy for Encrypted Files

- As discussed previously, entropy can also be used to detect encrypted files
 - E.g., In case the file signature is damaged, or modified
- For our earlier plaintext file, we can calculate the entropy as 4.49
 - This site will allow you to calculate entropy given a small number of bytes: <https://asecuritysite.com/encryption/ent>
 - Remember, calculating entropy for byte value gives a range from 0 to 8 (since there are 256 possible bytes)
 - This will be a common entropy value for other English language text

File Entropy for Encrypted Files

- For our encrypted ciphertext, we calculate the entropy as 6.05
- This tells us that the data is more random (uniformly distributed)
 - We expect this for encrypted data
- However, we must also recognise though that there are different ways to encode data
- With the previous openssl encryption, a ciphertext with binary output was produced (this is the default)

File Entropy for Encrypted Files

- Encrypted data, can also be encoded in other ways (the same is true for all data)
- For example, base64 encoding is a way to convert data that uses all 256 bytes, to data that only uses 64 printable characters
- For this variation, we can use the command:

```
$ openssl enc -aes256 -base64 -in plain -out ciphertextB64
```
- In this case, we can see the difference in the ciphertext outputs

File Entropy for Encrypted Files

- Default encryption with binary data

```
$ xxd ciphertext
00000000: 5361 6c74 6564 5f5f 8ae9 4b9b 0d2a 9228  Salted__..K..*..(
00000010: 82bf 4dc5 38ad 0c67 12a1 3ddf 991b ae25  ..M.8..g..=....%
00000020: 6c39 b349 085d 3958 95f1 eae5 8bed e7f9  l9.I.]9X.....
00000030: 7143 f5a8 b11a 0d8f 19f2 f819 4f86 4e1c  qC.....O.N.
00000040: 4582 c8f0 ad53 3e5b ebd2 dfe2 40b3 2ac3  E....S>[....@.*.
```

- Base64 encrypted file

```
$ xxd ciphertextB64
00000000: 5532 4673 6447 566b 5831 3931 444b 6439  U2FsdGVkX191DKd9
00000010: 4538 3876 7562 5766 7274 4b52 5857 3077  E88vubWfrtKRXW0w
00000020: 4862 6f6f 6732 6978 5771 506f 5372 7950  Hboog2ixWqPoSryP
00000030: 6c4c 5553 7379 5176 3945 7376 6a7a 7a36  lLUSsyQv9Esvjzz6
00000040: 0a6d 4751 7039 7639 6672 3173 4f74 6c75  .mGQp9v9fr1sOtlu
00000050: 5543 6856 3370 3044 7734 456f 3336 4777  UChV3p0Dw4Eo36Gw
00000060: 6476 5037 4f46 3564 4c4f 4173 3d0a      dvP7OF5dLOAs=.
```

- Both ciphertext files are encrypted with AES256 with the same key, but the ciphertext files are just encoded differently
 - With the second file, the file content is evident with the ascii text in the right column

File Entropy for Encrypted Files

- In terms of entropy, we obtain the following:
 - plain: 4.49
 - ciphertext: 6.05
 - ciphertextB64: 5.53
- With the base64 encoding, which uses a smaller set of bytes, the entropy is still more random than the original plaintext
- However, the difference isn't as great as with the default ciphertext
- Thus, while encrypted data can be detected it can sometimes be tricky to recognize it, at least based on different encodings that can be used

File Entropy for Encrypted Files

- Note that for the two ciphertext files, the file signatures also differ
- For the default symmetric encryption (“ciphertext”), the file signature “53 61 6c 74 65 64 5f 5f” is used (as shown earlier)
 - Corresponding to the ascii text “Salted__”
- For the base64 encrypted file (“ciphertextB64”), the file signature “55 32 46 73 64 47 56 6b 58 31” is used
 - Corresponding to the ascii text “U2FsdGVkX1”
- Thus, a DF examiner searching for encrypted data can search based on file signatures (covering a variety of encryption applications and encodings) and file entropy

Today's topic overview

- Topic 5: Anti-Forensics & Cryptography
 - Introduction
 - Steganography
 - Cryptography
 - Implications for Digital Forensics

Next topic

- Topic 6: Forensic Imaging and Data Analysis
 - Incident Response Management
 - Forensic Imaging
 - Forensic Data Analysis